

SeongRae Noh

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RESEARCH SUMMARY

My research focuses on LLM agents, 3D scenes, and decision-making in interactive environments: LLM-based multi-agent cooperation in long-horizon and partially observable settings, decision-making under uncertainty, and goal-directed planning for verifiable actions in 3D scenes.

TECHNICAL SKILLS

Agentic AI & LLMs: LLM Agents, Multi-Agent Systems, POMDPs, LLM Evaluation, HuggingFace Transformers

ML & Embodied AI: PyTorch, vLLM, ThreeDWorld, VirtualHome, Unreal Engine 5

Tools & Languages: Python, C++, Docker, Git; Korean (Native), English (Advanced), Japanese (Intermediate)

RESEARCH EXPERIENCE

Graduate Researcher

Mar. 2024 – Present

Korea University, IIIXR Lab (Advisor: Prof. HyeongYeop Kang)

Seoul, South Korea

- Conducting research on LLM agents, embodied AI, multimodal planning, and agent evaluation in interactive 3D environments.
- Developed agentic planning methods for open-vocabulary 3D scene editing and uncertainty-aware multi-agent decision-making.

Undergraduate Researcher

Dec. 2021 – Feb. 2024

KyungHee University, IIIXR Lab (Advisor: Prof. HyeongYeop Kang)

Yongin, South Korea

- Designed and implemented an embodied LLM agent framework in VirtualHome, enabling multi-agent task planning, communication, and action validation for long-horizon household tasks.
- Developed generative 3D environment pipelines for agent simulation, creating urban layouts and Unreal Engine scenes for interactive embodied AI experiments.

INDUSTRIAL PROJECTS

Development of a Multimodal Information-Driven Service Agent Framework (LG Electronics) Oct. 2025 – Sep. 2026

- Designed a multimodal agent architecture for HVAC service workflows, combining LLM reasoning, visual understanding, and structured device data for context-aware diagnosis and action recommendation.

Understanding Multimodal PDF Documents Using LLMs (LG Electronics)

Jun. 2025 – Sep. 2025

- Built an LVLM-based document intelligence system for multimodal technical PDFs, enabling structured extraction, table/figure understanding, and cross-modal reasoning over service manuals.

SELECTED PUBLICATIONS

(SIGGRAPH Asia 2026) CuCaT: Cut-Structured Camera Trajectory Generation for Dance Music Videos

SooBin Lim*, **SeongRae Noh***, Jeongyoun Kwon, YoungBeom Kim, HyeongYeop Kang[†]

- Proposed a cut-structured camera formulation that separates cut and transition events over a performer-relative shot plan, achieving the best continuity and variation on the DCM benchmark while keeping the dancer visible in nearly every frame.

(NeurIPS 2026) Baba in Wonderland: Online Self-Supervised Dynamics Discovery for Executable World Models

SeungWon Seo*, DongHeun Han*, **SeongRae Noh**, HyeongYeop Kang[†]

- Proposed Alice, which learns executable world models online by turning rejected LLM program updates into hypothesis classes that guide both updates and exploration, reaching 0.97 balanced transition accuracy vs. below 0.20 for the closest baseline.

(CVPR 2026) Edit-As-Act: Goal-Regressive Planning for Open-Vocabulary 3D Indoor Scene Editing

SeongRae Noh, SeungWon Seo, Gyeong-Moon Park[†], HyeongYeop Kang[†]

- Formulated open-vocabulary 3D scene editing as agentic goal-regressive planning over world states, and introduced the E2A Benchmark, improving instruction fidelity by 20% and semantic consistency by 43.1%.

(ICLR 2026) From Assumptions to Actions: Turning LLM Reasoning into Uncertainty-Aware Planning for Embodied Agents

SeungWon Seo*, SooBin Lim*, **SeongRae Noh***, Haneul Kim, HyeongYeop Kang[†]

- Proposed uncertainty-aware scenario tree planning for LLM agents under partial observability, converting latent assumptions into robust multi-agent decisions while reducing communication by 71.1% and improving task performance by 8.6%.

(AAAI 2025 Oral) REVECA: Adaptive Planning and Trajectory-Based Validation in Cooperative Language Agents Using Information Relevance and Relative Proximity

SeungWon Seo*, **SeongRae Noh***, Junhyeok Lee, SooBin Lim, Won Hee Lee, HyeongYeop Kang[†]

- Developed a cooperative LLM-agent planning framework with relevance-filtered memory and trajectory-based validation, improving performance by 47% in noisy environments and user preference by 73%.

EDUCATION

Korea University <i>Integrated MS-Ph.D. in Computer Science, IIIXR LAB (Advisor: Prof. HyeongYeop Kang)</i>	Seoul, South Korea <i>Mar. 2024 – Feb. 2029</i>
KyungHee University <i>B.S. in Computer Science</i>	Yongin, South Korea <i>Mar. 2020 – Feb. 2024</i>

HONORS & SERVICE

Awards: Ph.D. Fellowship, National Research Foundation of Korea (2026, \$37,000); NeurIPS Top Reviewer (2026); AAAI Student Scholarship (2025); Teaching Assistance Scholarship (2024); Pearl Abyss Fellowship (2023, 2021); 2nd place, KHU Software Convergence Inter-Club Competition (2022)
Service: Reviewer – ICLR (2027), NeurIPS, ICML (2026), ACM MM, Scientific Reports (2025); Student Volunteer – SIGGRAPH Asia (2026), AAAI (2025)